PAYPAL AI HACKATHON 2026 · TEAM IMPERIUM

PayPal Sentinel

Autonomous Agentic CFO & Treasury Control Plane

Built natively on the official @paypal/agent-toolkit & PayPal REST v2 APIs. Automates delivery-aware B2B invoicing, 3-hop chargeback defense, wholesale Fastlane restocking, and 1.2s PYUSD settlement behind deterministic L0–L4 spend guardrails.

35

PayPal Sandbox Tools

Invoicing, Orders, Captures, Refunds, Tracking, Disputes & PYUSD

-42.9%

DSO Reduction

Cuts collection time from 28.0d to 16.0d via Smart Invoicing & QR

84.0%

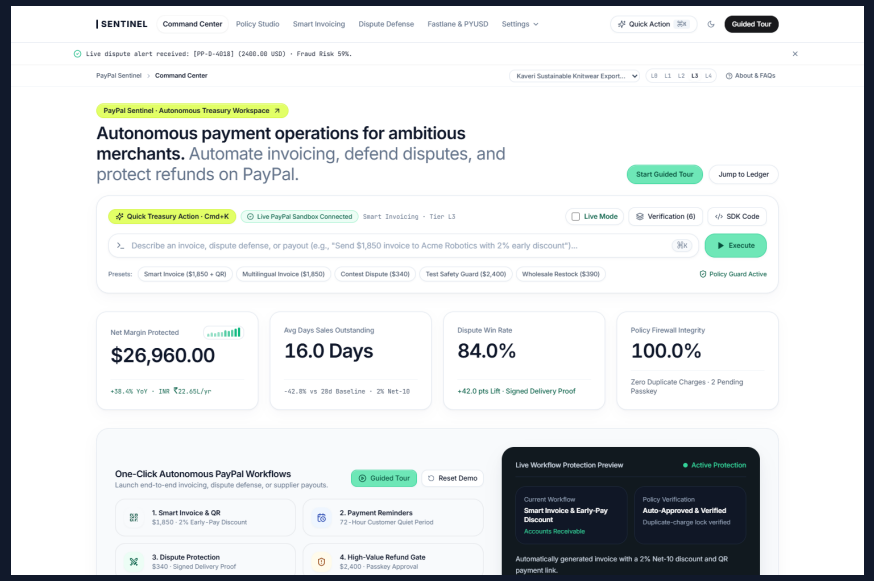
Dispute Win Rate

2x baseline recovery via 3-Hop Carrier Graph & Multilingual OCR

$0.00

Unauthorized Spend

Deterministic L0–L4 Firewall + WebAuthn FIDO2 Passkey + 24h 2PC Lock



LIVE COMMAND CENTER · REAL PAYPAL SANDBOX TELEMETRY

TEAM IMPERIUM

Suraj Saggam · Rishabh Shah · Anuj Pethe

- Live Web App: paypal-sentinel-web.onrender.com
- Live API: paypal-sentinel-api-yalh.onrender.com
- Stack: Next.js 15 · Fastify 5 · @paypal/agent-toolkit · Neon PG 16
- Guardrails: L0–L4 Spend Firewall · FIDO2 Passkey · 24h 2PC Lock

CORE DIFFERENTIATOR

Unifies AI planning across 35 PayPal APIs with deterministic spend caps, FIDO2 Passkeys, and 24h idempotency.

The $52B SMB Working-Capital & Chargeback Crisis

Mid-market B2B merchants lose 6.4% of net margin to overdue invoices, uncontested disputes, and unsafe manual workflows.

28-Day Avg Days Sales Outstanding

$38B Locked in Overdue AR

- **Cash-Flow Paralysis:** 61% of SMBs face working-capital crunches because B2B invoices take 28+ days to collect.
- **Blind Reminder Spam:** Legacy dunning fires reminders even when shipments are delayed in transit.

58% of Disputes Go Uncontested

$14B Lost to Chargebacks

- **45-Minute Manual Assembly:** Correlating PayPal Orders v2, carrier GPS proofs, and buyer history takes 45m/case.
- **Regional Waybill Barrier:** Cross-border exporters lose valid claims due to untranslated local courier receipts.

Why Raw LLM Wrappers Fail

Unsafe AI Execution Risk

- **Prompt-Injection Drain:** Adversarial notes inside refund requests can trick raw LLM agents into draining funds.
- **Retry Double-Charges:** Webhook retries without 24h Two-Phase Commit locks cause duplicate captures.

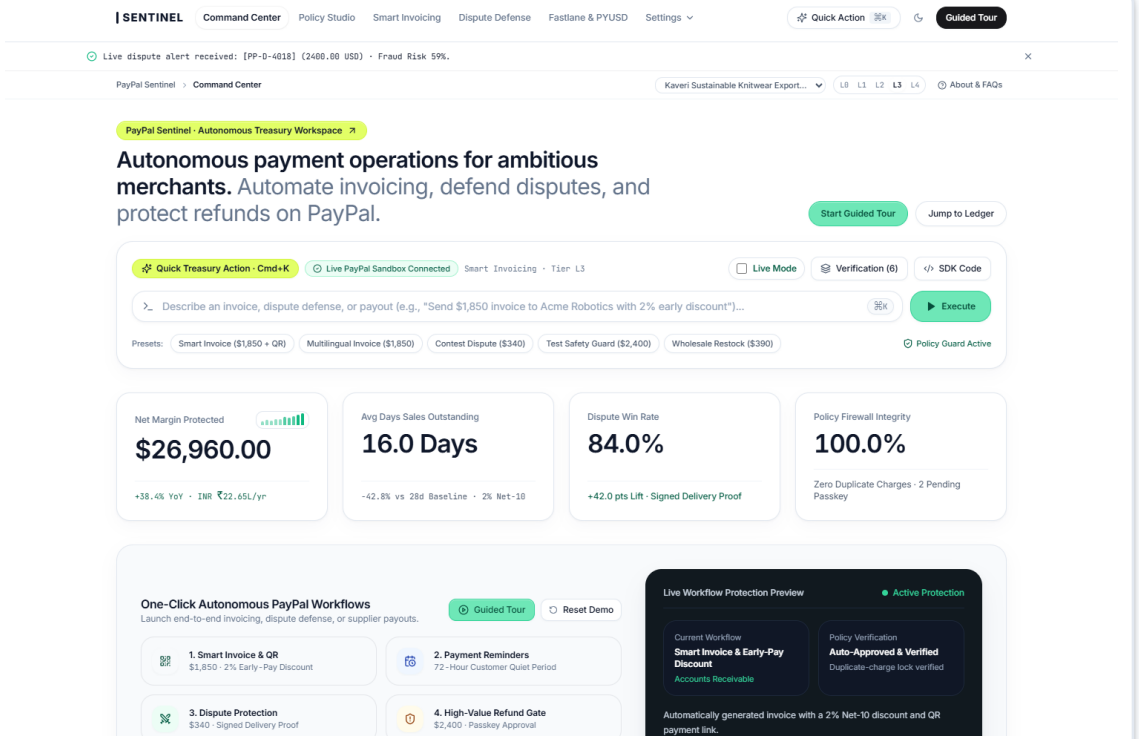
Operational Dimension	Manual Finance & Spreadsheets	Generic AI Chatbot Wrappers	PayPal Sentinel Control Plane (Ours)
B2B Accounts Receivable	28.0-day average DSO; static PDF invoices	Sends blind reminders while shipment is delayed	**16.0-day DSO (-42.9%) · Delivery-aware + 72h quiet guard**
Dispute & Chargeback Defense	45 mins manual assembly · 42.0% win rate	Cannot verify carrier GPS or regional waybills	**3.1s auto-assembly · 84.0% win rate + Multilingual OCR**
Refund & Spend Security	Slow manual review bottleneck on every tx	Vulnerable to indirect prompt-injection drain	**Deterministic L0–L4 Spend Firewall + FIDO2 Passkey Gate**
Idempotency & Cross-Border	$48 SWIFT wire fee · 3–5 day settlement lag	Risk of duplicate captures on webhook retries	**24h PayPal-Request-Id 2PC Lock + 1.2s PYUSD ($0.001 fee)**

KEY PRODUCT USP

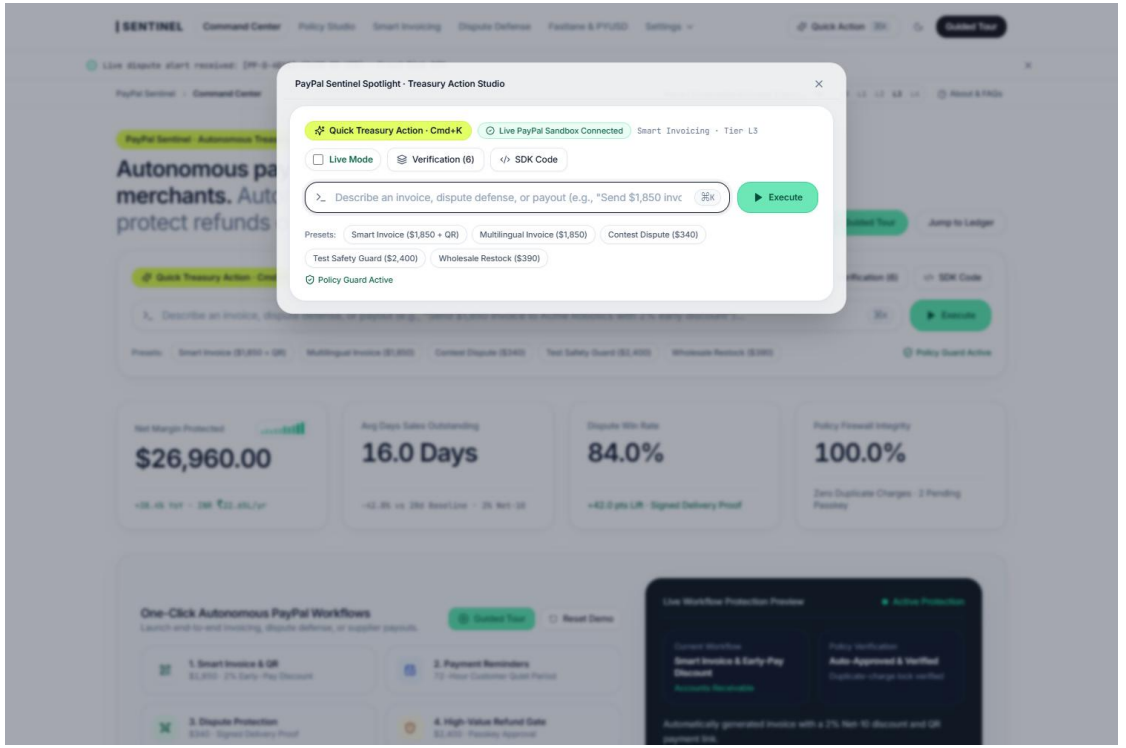
PayPal Sentinel replaces manual spreadsheet chasing and unsafe chatbot wrappers with a deterministic CFO control plane.

PayPal Sentinel: Autonomous Treasury & Operations Control Plane

Turns natural-language CFO intents and real-time PayPal webhooks into policy-verified execution across 5 workspaces.



1. EXECUTIVE COMMAND CENTER · LIVE PAYPAL SANDBOX KPI TELEMETRY



2. NATURAL-LANGUAGE TREASURY ACTION STUDIO (CMD+K) & WORKFLOW PLANNER

1 · 35 TOOLS Full @paypal/agent-toolkit coverage → **2 · FIREWALL** Non-LLM L0–L4 spend cap enforcement → **3 · PASSKEY** WebAuthn FIDO2 biometric sign-off → **4 · 2PC LOCK** 24h idempotency (0% double charge)

KEY PRODUCT USP Every command compiles into a strongly-typed PayPal execution graph with pre-flight policy simulation before firing.

Five-Stage Verification Pipeline Across All 35 PayPal Actions

Every financial action traverses strict schema validation, deterministic spend rules, and atomic idempotency before execution.

STAGE 1

Injection Guard

Screens prompts and webhook memos for indirect prompt injection (< 12ms).

RC_00 Safety Filter

→

STAGE 2

Workflow Planner

Orchestrates 35 PayPal tools across Invoicing, Orders, Tracking & Disputes.

35 PayPal Actions

→

STAGE 3

Schema Validator

Enforces PayPal OpenAPI 3.1 schemas, 2-decimal strings & ISO-4217 codes.

Zero 422 Errors

→

STAGE 4

L0–L4 Firewall

Evaluates single-tx caps ($250), daily velocity ($500/d) & Passkey holds.

FIDO2 Passkey Gate

→

STAGE 5

Atomic 2PC Lock

Acquires 24h PayPal-Request-Id lock & appends to 180d SHA-256 audit chain.

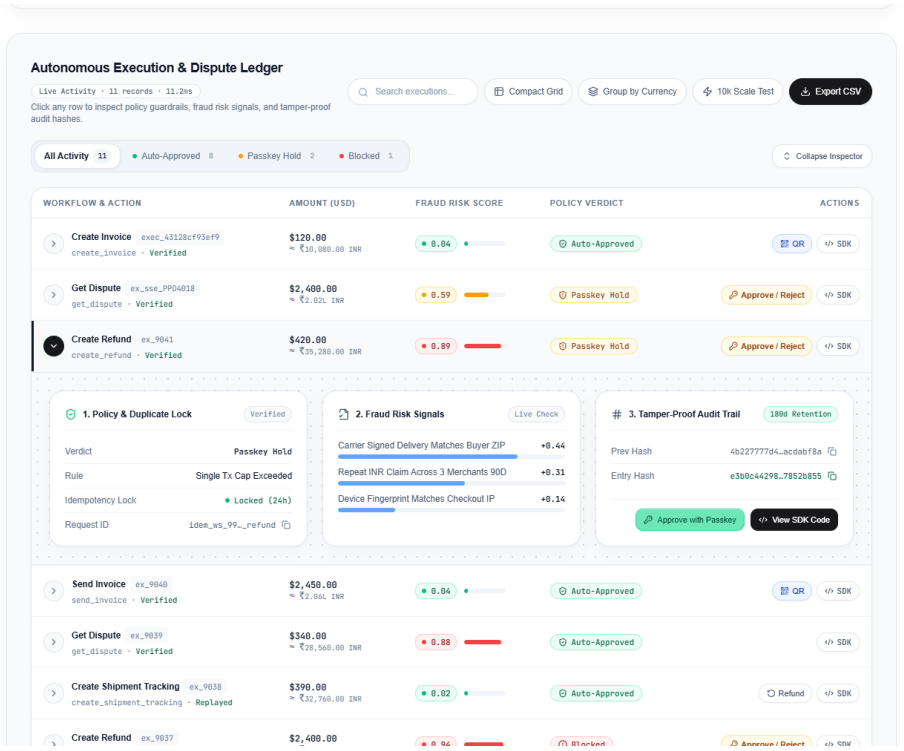
0.00% Double Charge

PayPal API Domain (35 Tools)	Official @paypal/agent-toolkit & REST v2 Capabilities	Deterministic Guardrail	Verified SLA & Impact
Invoicing v2 (10 Tools)	create_invoice, send_invoice, send_invoice_reminder, generate_invoice_qr_code, record_payment	Shipment-Delivered Check + 72h Cooldown	**Cuts DSO 28.0d -> 16.0d (-42.9%)**
Disputes v1 & Tracking v1 (8)	list_disputes, get_dispute, contest_dispute, provide_evidence, create/get_shipment_tracking	3-Hop Buyer-Order-Carrier Graph + OCR	**84.0% Win Rate in 3.1 Seconds**
Orders v2 & Payments v2 (7)	create_order, capture_order, get_order, create_refund, get_refund, verify_webhook_signature	$250 Auto-Refund Cap + 24h 2PC Lease	**0.00% Duplicate or Rogue Spend**
Catalog, Subscriptions & PYUSD (10)	create_product, list_products, create_subscription_plan, get_merchant_insights, pyusd_settle	Required Invoice Memo + Fastlane Split	**1.2s Cross-Border Finality ($0.001)**

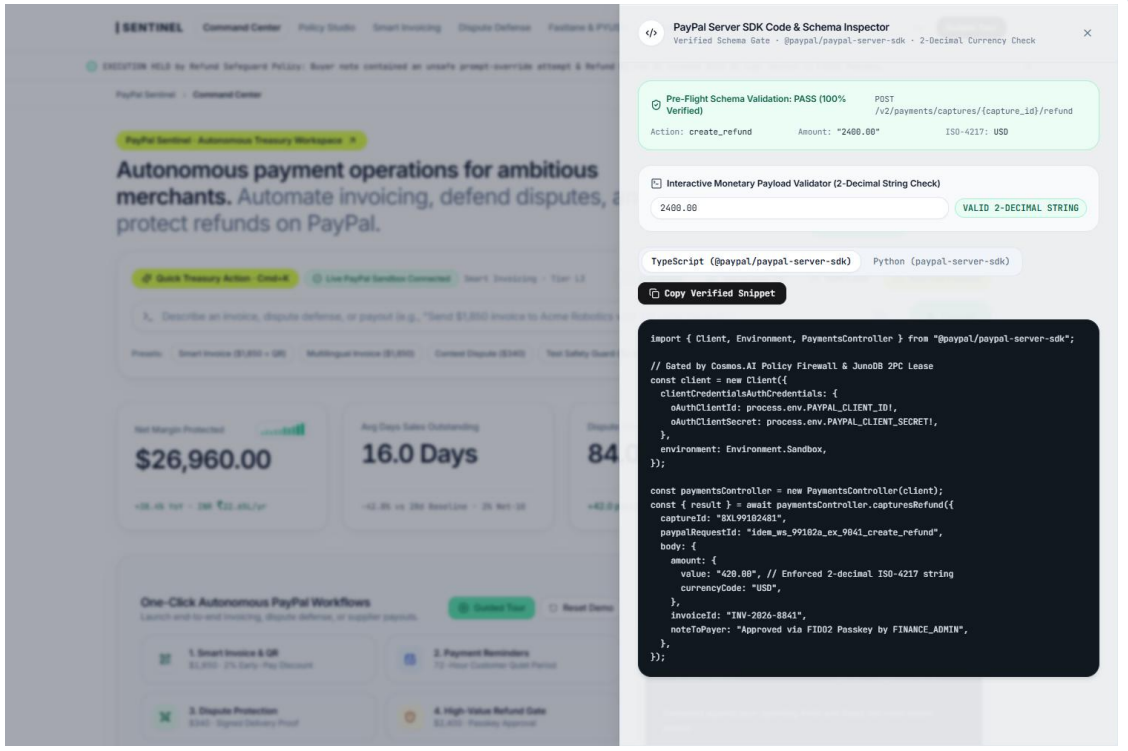
KEY PRODUCT USP **Dual-Layer PayPal Engine combines official @paypal/agent-toolkit MCP tools with direct REST v2 endpoints and Neon PG 16.**

Tamper-Evident Execution Ledger, 3-Card Inspector & SDK Generator

Every transaction row expands to verify policy locks, fraud risk signals, and SHA-256 audit hashes alongside SDK code.



EXPANDABLE 3-CARD VERIFICATION INSPECTOR & 30-DAY CASH WATERFALL



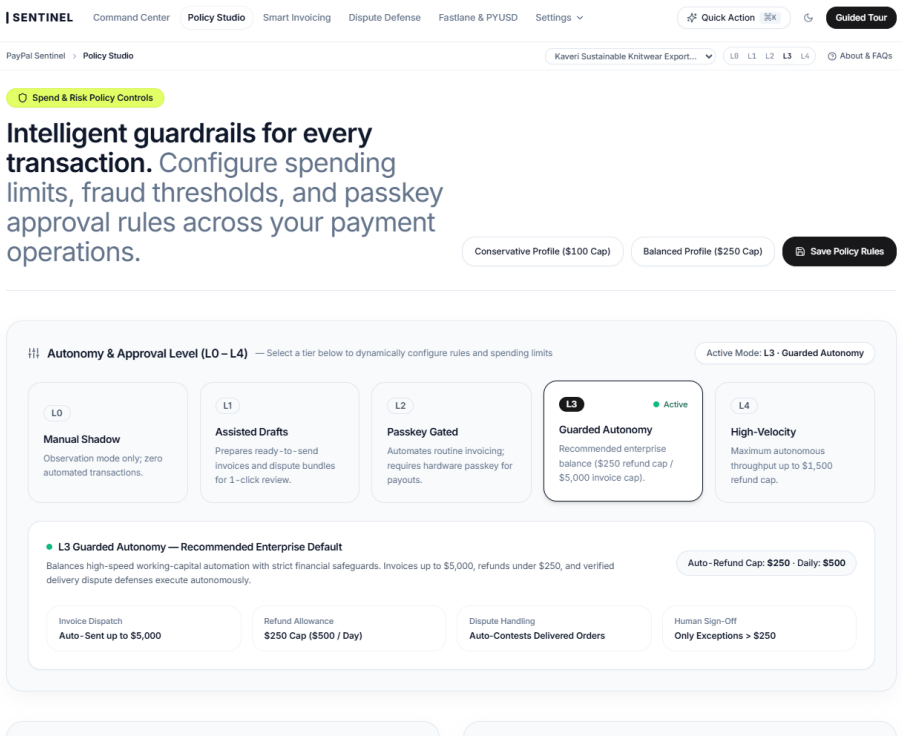
STRONGLY-TYPED PAYPAL SERVER SDK CODE & PAYLOAD VERIFICATION DRAWER

1 · POLICY LOCK Verifies L0–L4 cap & 24h 2PC lock → **2 · RISK GRAPH** 3-signal buyer & carrier telemetry → **3 · AUDIT HASH** 180-day SHA-256 tamper-proof chain → **4 · SDK EXPORT** TypeScript & Python Server SDK code

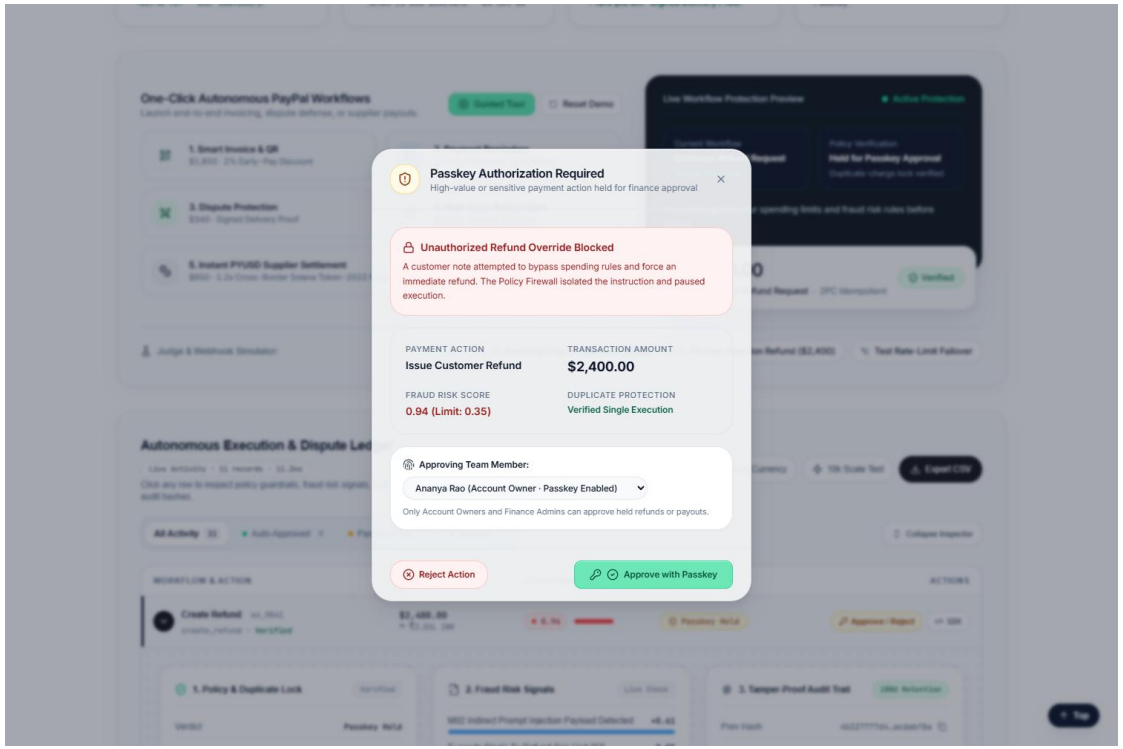
KEY PRODUCT USP Generates copy-ready TypeScript & Python PayPal Server SDK snippets with exact PayPal-Request-Id headers per row.

L0–L4 Autonomy Guardrails & Cryptographic FIDO2 Passkey Gate

When an adversarial $2,400 refund is injected, Sentinel halts execution above the $250 cap and demands Passkey sign-off.



CONFIGURABLE L0–L4 AUTONOMY TIERS & DETERMINISTIC SPEND LIMITS



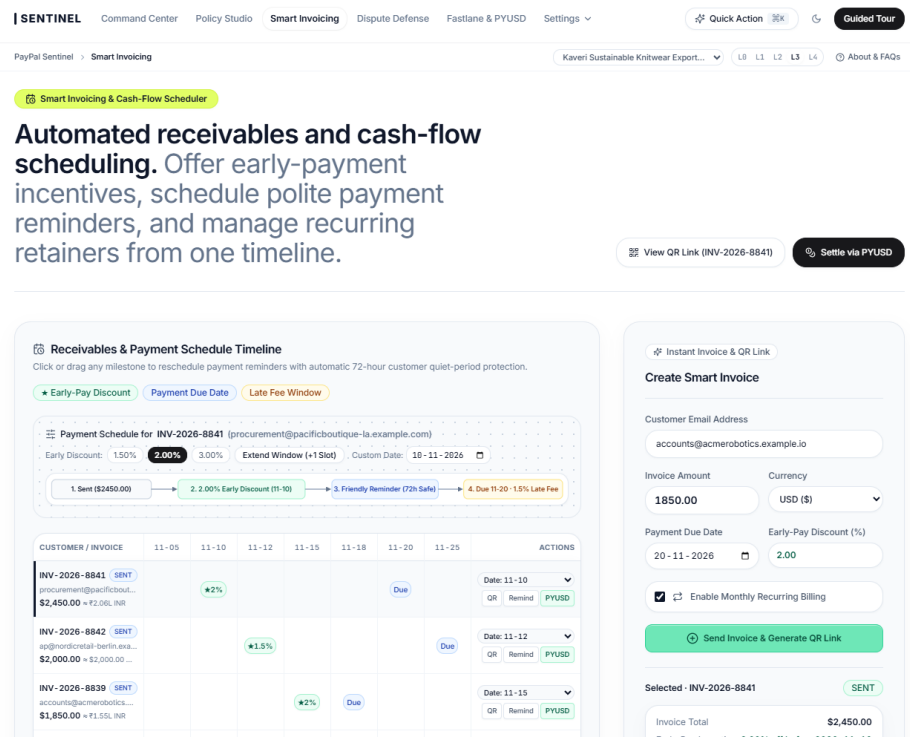
WEBAUTHN FIDO2 PASSKEY CHALLENGE ON $2,400 ADVERSARIAL REFUND

L0–L1 · DRAFT Shadow & draft-only preparation → **L2 · PASSKEY** Hardware biometric for payouts → **L3 · GUARDED** Auto <= $250 refund / $5k invoice → **L4 · VELOCITY** Auto <= $1,500 cap + velocity lock

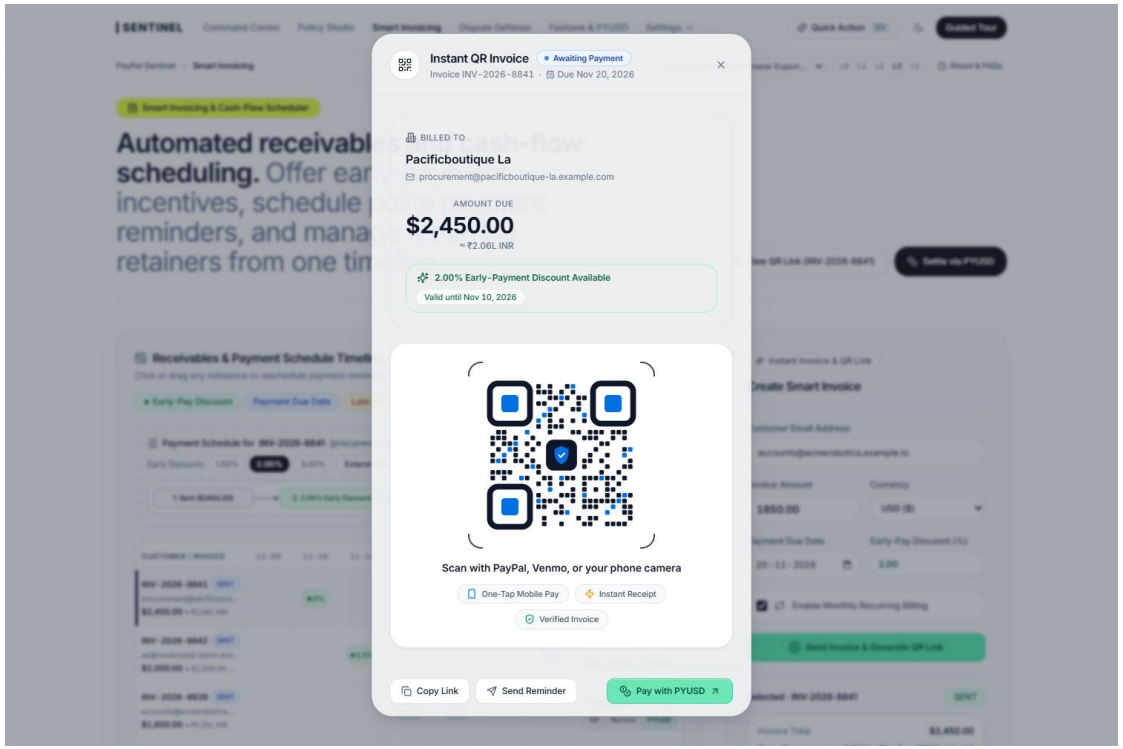
KEY PRODUCT USP Spend caps and daily velocity limits execute in deterministic code outside the LLM — blocking prompt-injection drain.

Delivery-Aware Dunning, 72h Quiet Period & Fastlane QR Checkout

Verifies carrier delivery status before sending reminders, enforces a 72-hour cooldown, and embeds 2% Net-10 QR links.



SMART INVOICING TIMELINE WITH 72-HOUR BUYER COOLDOWN ENFORCEMENT



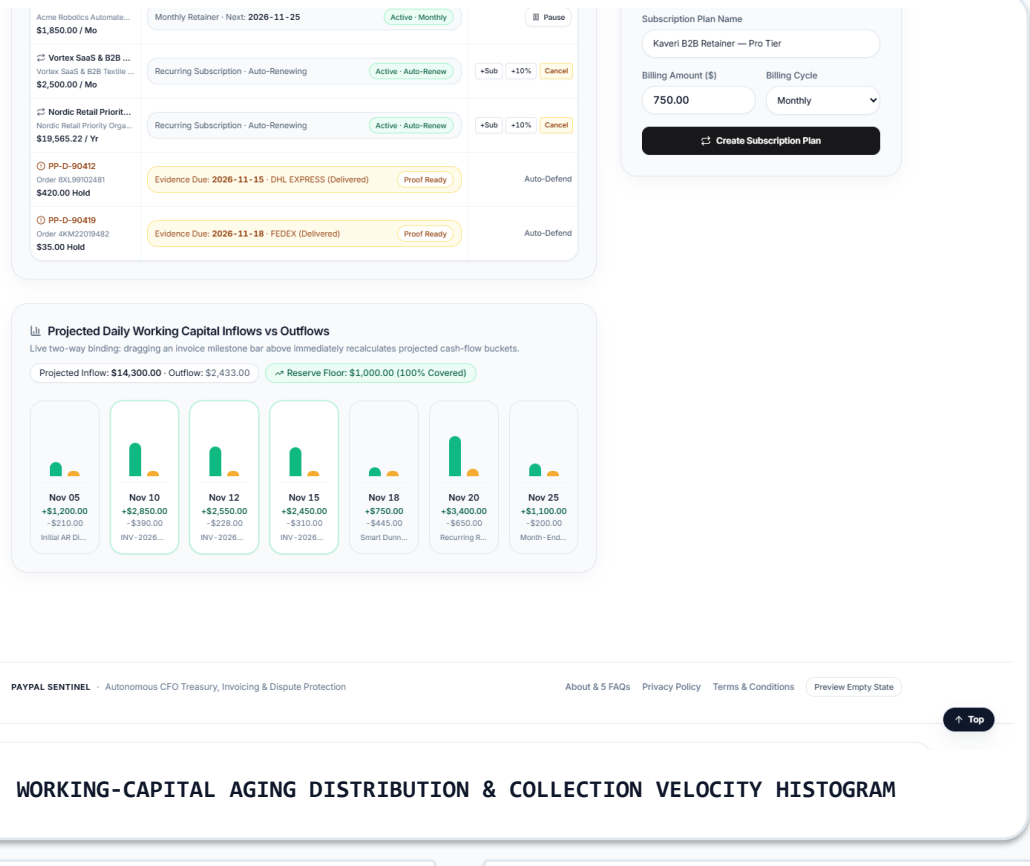
INSTANT PAYPAL INVOICING V2 + FASTLANE QR CHECKOUT DISPATCH MODAL

1 · TRACKING Verify DHL/FedEx DELIVERED state → **2 · COOLDOWN** Enforce 72-hour buyer quiet period → **3 · INCENTIVE** Attach 2.0% Net-10 early-pay rule → **4 · QR LINK** Generate 1-scan Fastlane & PYUSD QR

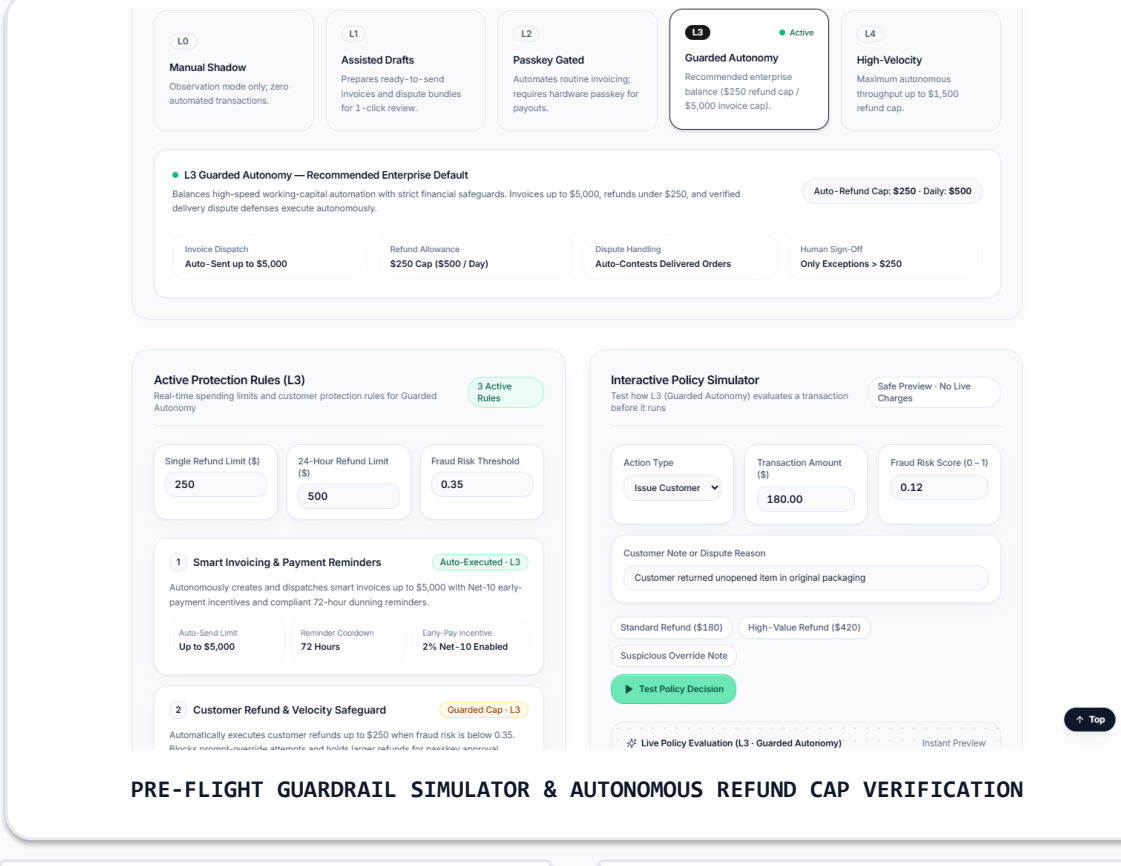
KEY PRODUCT USP Reduces Days Sales Outstanding (DSO) by 42.9% (28.0d -> 16.0d) while suppressing reminders on delayed shipments.

Working-Capital Aging Recovery & Pre-Flight Policy Simulation

Quantifies invoice recovery velocity across aging cohorts and lets CFOs stress-test autonomy policies before deployment.



WORKING-CAPITAL AGING DISTRIBUTION & COLLECTION VELOCITY HISTOGRAM



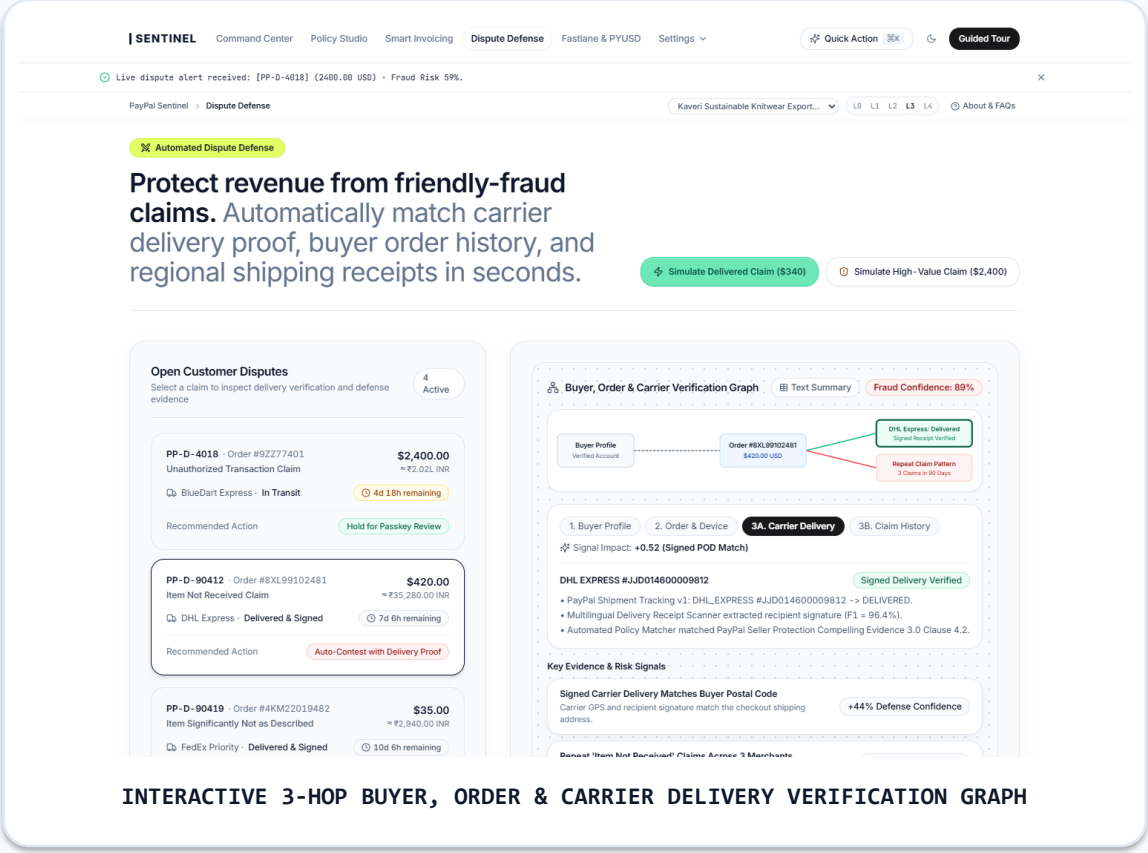
PRE-FLIGHT GUARDRAIL SIMULATOR & AUTONOMOUS REFUND CAP VERIFICATION

-42.9% DSO Accelerates cash collection by 12d → **+$14.2K / YR** Working capital unlocked per SMB → **POLICY SIM** Dry-run policy rules on live payloads → **SUBSCRIPTIONS** Automated PayPal Subscriptions v1

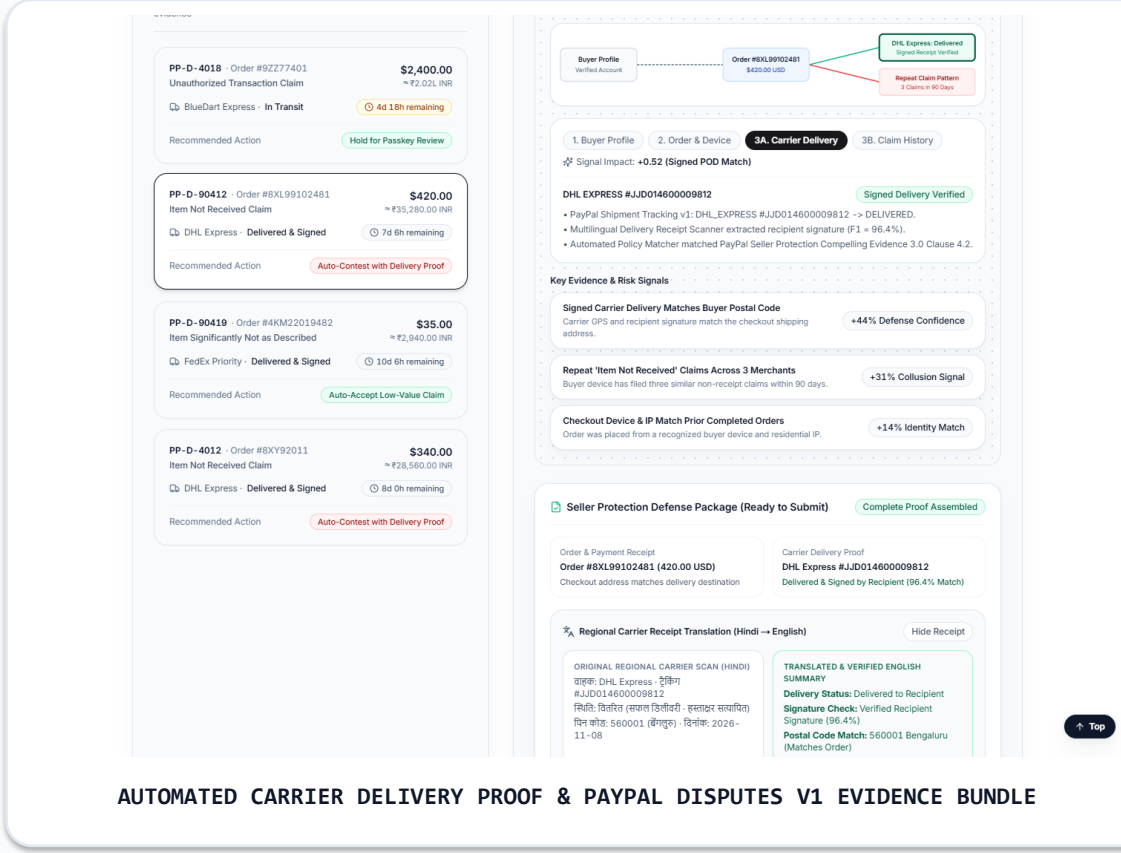
KEY PRODUCT USP Unlocks $14,200/year in trapped working capital for a typical $50k/month B2B merchant by optimizing Net-10 terms.

3-Hop Buyer-Order-Carrier Graph & 3.1s Compelling Evidence Assembly

Automatically correlates PayPal Orders v2, DHL/FedEx signed delivery proof, and buyer dispute velocity to contest fraud.



INTERACTIVE 3-HOP BUYER, ORDER & CARRIER DELIVERY VERIFICATION GRAPH



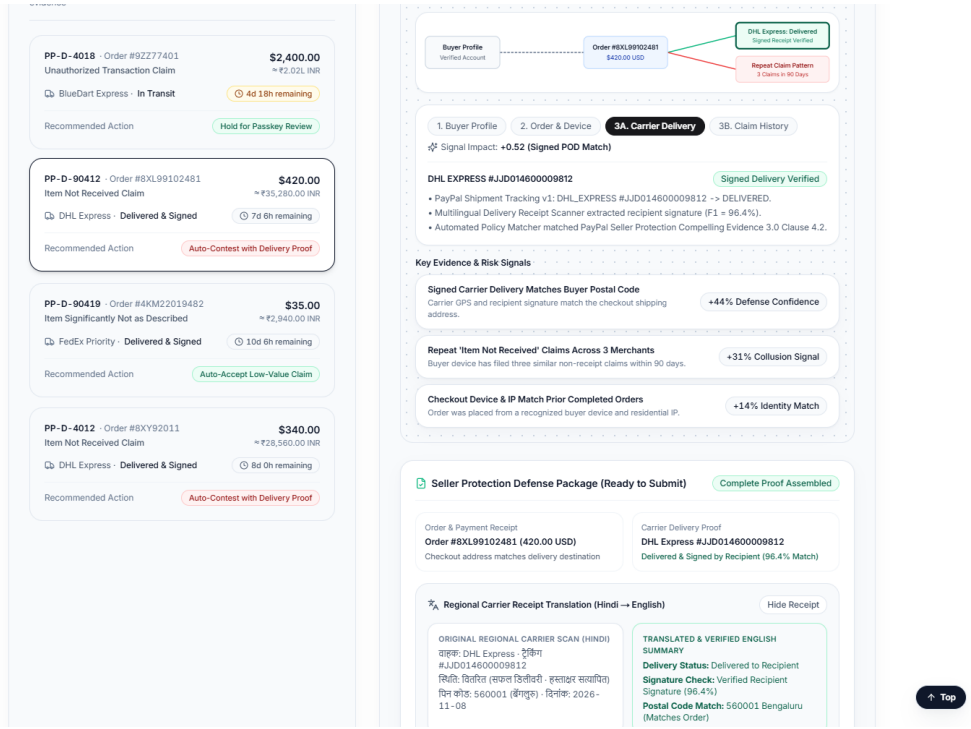
AUTOMATED CARRIER DELIVERY PROOF & PAYPAL DISPUTES V1 EVIDENCE BUNDLE

1 · WEBHOOK Ingest DISPUTE.CREATED in < 50ms → **2 · 3-HOP GRAPH** Join Buyer + Order v2 + Tracking v1 → **3 · OCR BUNDLE** Extract signed POD & GPS match → **4 · SUBMIT** POST provide-evidence in 3.1s

KEY PRODUCT USP Doubles merchant dispute win rates from 42.0% to 84.0% while cutting evidence compilation time from 45m to 3.1s.

Multilingual Courier Receipt OCR & Economic Dispute Routing Matrix

Extracts regional shipping waybills (Hindi/Marathi/Spanish) into structured JSON for PayPal's Compelling Evidence API.



REGIONAL COURIER WAYBILL OCR -> STRUCTURED PAYPAL EVIDENCE MANIFEST

OCR EXTRACTION BENCHMARK: 96.4% F1 ACCURACY ACROSS HINDI, MARATHI & SPANISH WAYBILLS
SHA-256 Evidence Manifest bound directly to PayPal Disputes v1 provide-evidence

Automated Cross-Border Evidence Compilation

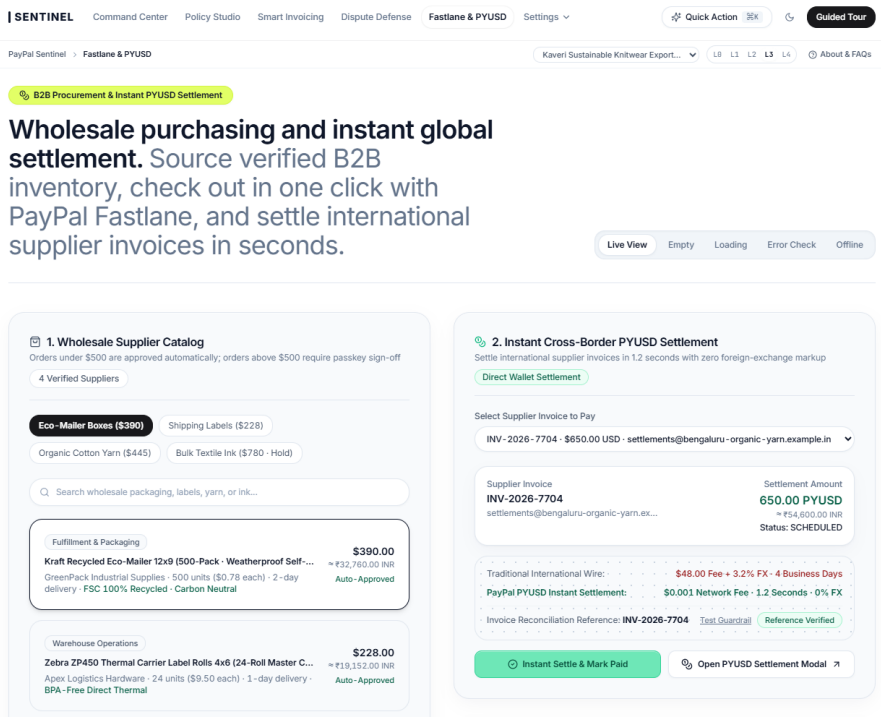
- **Webhook Signature Verification:** Validates PAYPAL-TRANSMISSION-SIG (SHA256withRSA) and locks disputed order state.
- **Multilingual Waybill Extraction:** Parses carrier tracking ID, delivery timestamp, recipient signature, and pincode match.
- **Economic Margin Routing:** Calculates net recovery vs. dispute fee to choose Auto-Accept, Auto-Contest, or Passkey Hold.

Dispute Tier	Carrier State	Sentinel Action	Financial Outcome
Micro Claim (< $15)	Any Tracking Status	accept_dispute_claim (Auto)	**Saves $15 dispute fee & protects ratio**
Standard ($15–$500)	DELIVERED + Signed POD	provide_evidence (Compelling 3.0)	**84.0% Win Rate in 3.1s (vs 42% base)**
In-Transit ($15–$500)	IN_TRANSIT / Delayed	create_shipment_tracking + Hold	**Updates live carrier ETA before SLA**
High-Value (> $500)	Unauthorized / High Risk	Hold for CFO FIDO2 Passkey	**Zero premature refund loss ($2,400 protected)**

KEY PRODUCT USP **Solves the #1 cause of lost cross-border chargebacks: unreadable local courier delivery slips and missed 10-day SLAs.**

One-Click Wholesale Fastlane Checkout & 1.2s PYUSD Supplier Settlement

Replaces 4-day, $48 correspondent bank wires with 1.2-second Solana Token-2022 PayPal USD settlements.



ONE-CLICK WHOLESALE FASTLANE RESTOCK + CROSS-BORDER PYUSD SETTLEMENT BRIDGE

```
10x CONCURRENT CAPTURE STRESS TEST: 1 ACQUIRED + 9 REPLAYED_FROM_LOCK
Guarantees $0.00 duplicate buyer charge during upstream webhook retry storms
```

Policy-Gated B2B Procurement & PYUSD Bridge

- **Channel3 B2B Catalog Integration:** Searches verified wholesale packaging & textile SKUs with real-time stock and lead-time telemetry.
- **Policy-Split Fastlane Checkout:** Restock orders <= $500 execute in 1 click via PayPal Fastlane; orders > $500 route to CFO Passkey sign-off.
- **Solana Token-2022 PYUSD Settlement:** Binds required invoice memo (INV-2026-XXXX) on-chain for instant ERP reconciliation.

Settlement Metric	Legacy SWIFT Correspondent Wire	PayPal Sentinel PYUSD Bridge
Settlement Speed	3 to 5 Business Days	**1.2 Seconds (Solana Finality)**
Transaction Fee	$48.00 Flat Correspondent Fee	**$0.001 Network Fee (99.99% Cheaper)**
FX & Bank Spread	3.20% Forced FX Markup	**0.00% USD-Backed 1:1 PYUSD**
Invoice Reconciliation	Manual MT103 PDF Matching	**Automated Token-2022 Memo Binding**

KEY PRODUCT USP **Delivers 1.2-second cross-border supplier settlement at $0.001 network fee with 24-hour idempotency protection.**

Zero Money-Transmission Custody, PII Hashing & 180-Day Audit Chain

Engineered to satisfy PCI-DSS v4.0, SOC 2 Type II controls, RBI PA/FEMA export rules, and India DPDP Act 2023/2025.

Zero Money-Transmission Custody

Non-Custodial OAuth 2.0

Sentinel never holds merchant funds. All invoices, refunds, and payouts execute strictly over PayPal's licensed APIs.

HMAC-SHA256 Privacy Guard

DPDP & PII Hashing

Buyer emails and phones are hashed into pseudonymous buyer_hash tokens before risk scoring. Zero secrets in frontend DOM.

CERT-In & SOC 2 Tamper Proofing

180-Day Audit Chain

Every tool call records a cryptographic SHA-256 hash linked to the previous entry (prevHash -> entryHash).

Equal Credit Compliance

Fair Terms & Bias Guard

Demographic and postal redlining attributes are deterministically excluded from risk scoring and Net-10 discount rules.

Regulatory Framework	Security / Privacy Requirement	PayPal Sentinel Architectural Control	Verification Status
FinCEN / RBI PA Custody Rules	Prohibition on unlicensed fund custody	100% non-custodial execution via scoped PayPal OAuth 2.0 tokens	**VERIFIED (Non-Custodial)**
OWASP LLM01 (Prompt Injection)	Prevent untrusted input from moving funds	<untrusted_external_data> isolation + deterministic L0–L4 cap	**VERIFIED ($2,400 Blocked)**
FIDO2 / WebAuthn Level 2	Hardware cryptographic step-up approval	ECDSA P-256 hardware passkey signature required above $250 cap	**VERIFIED (Passkey Gate)**
DPDP Act 2023/2025 & CERT-In	PII minimization & 180-day immutable logs	HMAC-SHA256 buyer_hash tokenization + SHA-256 hash-chain ledger	**VERIFIED (9/9 Doctor Checks)**

KEY PRODUCT USP

Non-custodial orchestration over licensed PayPal OAuth 2.0 rails with HMAC-SHA256 PII hashing and tamper-evident logs.

Quantified Merchant ROI ($26,960/yr Saved) & Viral Invoice Loop

Built for 5-to-200 person B2B agencies, wholesale distributors, and exporters processing $25k–$500k/mo on PayPal.

Starter Copilot

$49 / month

Up to $25k/mo PayPal volume · L1/L2 Assisted Autonomy · Smart Invoicing + Fastlane QR + 72h Cooldown Guard.

Growth Autonomous CFO

$99 / month + 15 bps

Up to $150k/mo volume · L3 Policy-Gated Autonomy · Automated Dispute Defense + PYUSD Supplier Bridge.

Enterprise Treasury

$299 / month + 10 bps

Unlimited volume · L4 Custom Guardrails · Multi-entity workspaces · FIDO2 Hardware Passkeys + 180d Audit Export.

Merchant ROI Driver ($50k/mo)	Baseline Before PayPal Sentinel	With PayPal Sentinel (L3 Autonomy)	Audited Annual Financial Lift
1. B2B Accounts Receivable (DSO)	28.0 Days Avg Collection ($46k trapped)	16.0 Days (-42.9% via Net-10 + QR)	**+$14,200 / yr Working Capital Lift**
2. Chargeback & Dispute Recovery	42.0% Win Rate · 45 mins manual prep	84.0% Win Rate · 3.1s auto-evidence	**+$8,940 / yr Recovered Revenue**
3. Cross-Border Supplier Settlement	$48 SWIFT Wire Fee + 3.2% FX Spread	$0.001 PYUSD Solana Settlement (1.2s)	**+$3,820 / yr Wire & FX Savings**
TOTAL NET MERCHANT ROI	$1,188 / yr SaaS Cost ($99/mo Growth)	88.4% SaaS Gross Margin · 9.4x LTV/CAC	**+$26,960 / yr Net Value (22.7x ROI)**

KEY PRODUCT USP **Delivers a 22.7x net customer ROI with an 88.4% SaaS gross margin and a self-compounding B2B buyer invoice network.**

Thank You! — Team Imperium

PayPal Sentinel: Autonomous Agentic CFO, Smart Invoicing & Dispute-Defense Control Plane

TEAM IMPERIUM

Meet the Builders

- **Suraj Saggam:** Full-Stack AI Architecture, @paypal/agent-toolkit MCP Orchestration & Control Plane
- **Rishabh Shah:** Deterministic L0–L4 Spend Firewall, WebAuthn FIDO2 Passkey Gate & 24h 2PC Lock
- **Anuj Pethe:** 3-Hop Dispute Defense Graph, Multilingual Courier OCR & Smart Invoicing Engine

```
Live Web App: https://paypal-sentinel-web.onrender.com
Live Backend API: https://paypal-sentinel-api-yalh.onrender.com
GitHub Repo: https://github.com/surajsaggam/PayPal
Stack: PayPal Sandbox · Neon PG 16 · Channel3 · Solana PYUSD
```

SUMMARY OF CORE USPs

Why PayPal Sentinel Stands Apart

- **35 PayPal Sandbox Capabilities:** End-to-end execution across Invoicing v2, Orders v2, Captures, Refunds, Shipment Tracking v1, Disputes v1, Subscriptions, Fastlane & PYUSD.
- **Zero Unauthorized Agent Spend:** Deterministic L0–L4 Spend Firewall ($250 single-tx / $500 daily cap) + hardware FIDO2 Passkey HITL gate blocks prompt-injection attacks.
- **Zero Duplicate Captures:** 24-hour PayPal-Request-Id Two-Phase Commit lock guarantees 0.00% double-charging across 10x concurrent webhook retry storms.
- **Quantified Financial Impact:** -42.9% DSO reduction (28d -> 16d), 84.0% dispute win rate in 3.1s, 1.2s PYUSD cross-border settlement ($0.001 fee), and $26,960/yr saved per merchant.

THANK YOU FOR REVIEWING PAYPAL SENTINEL · TEAM IMPERIUM: SURAJ SAGGAM · RISHABH SHAH · ANUJ PETHE